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Measures of Dispersion

Introduction

According to the 2006 General Social Survey, Americans have completed an average of 13.29 years of education. Of course this does not mean that each and every American has exactly 13.29 years; some have more and some have less. The 2006 General Social Survey also tells us that 15.3% of the U.S. population has less than a high school degree and that 16.9% has a bachelor's degree; but these ordinal measures tell us very little about how cases are distributed around the average of 13.29 years. Do most

Dispersion The degree to which cases are clustered around a particular value, usually the mean.

Americans have just about 13–14 years? Or are there lots of people with 10 years and lots of people with 16 years?

These are questions that measures of central tendency cannot help us to answer. They are questions that go beyond averages. In other words, once we know an average, we may want to know something more about the data it begins to describe. For example, the numbers 4, 5, and 6 have an average of 5. So do the numbers 3, 5, and 7. And so do the numbers 2, 5, and 8. You can see from these sets of numbers that an average of 5 can come from a group of tightly clustered numbers or from a group of widely dispersed numbers. Measures of dispersion tell us about the degree of clustering among cases in a data set. This degree of clustering is referred to as dispersion. The terms variability and dispersion are often used interchangeably, but the term dispersion will be used here. Table 1 presents relevant formulas and symbols for our discussion of dispersion.

TABLE 1 *Symbols and Formulas*

TERM/SYMBOL	MEANING OF SYMBOL	FORMULA
Measures of dispersion	A group of statistics (range, variance, standard deviation) that is used to describe how spread out cases are across a distribution. They tell us how tightly cases tend to cluster. Used with interval/ratio variables.	
Range	The distance between the case with the highest value and the case with the lowest value	$R = H - L$
Interquartile range	The distance from the third to the first quartile	$Q = Q_3 - Q_1$
Deviation	The distance that a particular case falls from the mean	$D_i = X - \bar{X}$
Mean deviation	The average distance that each case (measured as a positive value) falls from the mean	$D_t = \frac{\Sigma(\lvert X - \bar{X} \rvert)}{N}$
Sum of squares	The sum of squared deviations from the mean	$ss = \sum (X - \bar{X})^2$
Variance	A type of average. The average of the squared "distances" from the mean.	$s^2 = \frac{\Sigma(X - \bar{X})^2}{N}$
Standard deviation	A type of average equal to the square root of the variance. It is the square root of the average of the squared "distances" from the mean.	$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}}$

Many questions cannot be answered using measures of central tendency because measures of central tendency only tell us around what values cases tend to cluster. They do not tell us how tightly cases tend to cluster around those values. Therefore, we need a new group of statistics that describe the degree to which cases are clustered (or spread out). These statistics are known as measures of dispersion. Distributions can be described more completely when both measures of central tendency and measures of dispersion are used.

In this chapter we learn how to calculate and interpret five of the most commonly used measures of dispersion: the range, interquartile range, mean deviation, variance, and standard deviation. As in the case of measures of central tendency, it is important to understand what these measures tell us and what their limitations are. We learn how to calculate each of these with both raw and grouped data.

The Distribution of Education in the United States

The United States is a relatively highly educated country. According to data from the General Social Survey (National Opinion Research Center, 2006), Table 2, about 85% of Americans have at least a high school education and 34% of Americans have more than a high school education. This means that only about 15% of respondents have less than a high school degree. For these same respondents, 38% have fathers with less than a high school degree and 34% have mothers with less than a high school degree. Additionally, the mean number of years of education for all Americans is 13.29, compared to only 11.27 years for their mothers and 11.28 years for their fathers. This appears to

TABLE 2 *RS Highest Degree*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	LT HIGH SCHOOL	691	15.3	15.3	15.3
	HIGH SCHOOL	2273	50.4	50.4	65.8
	JUNIOR COLLEGE	377	8.4	8.4	74.1
	BACHELOR	763	16.9	16.9	91.1
	GRADUATE	403	8.9	8.9	100.0
	Total	4507	99.9	100.0	
Missing	NA	3	.1		
Total		4510	100.0		

indicate that Americans are an increasingly educated population. The United Nations *Human Development Report* ranks the United States 13th in the world on the *Human Development Index* (2009). This relatively high ranking is due in part to a strong record of public education.

While the United States is one of the most highly educated countries in world, this education is not distributed evenly across the population. For example, when we look at the distribution of respondents' highest degree by race, we find that 13% of Whites have less than 12 years of education, however, 21.6% of Blacks and 36.5% of Others have less than 12 years of education. These patterns are shown in Table 3 and in Figure 1. Clearly, it is important to look closely at these data to describe the distribution of education in American society. Because averages and percentages are not capable of fully describing a distribution, it is important to have an additional set of tools at our disposal. The set of tools to which we now turn our attention is measures of dispersion. We begin our investigation of measures of dispersion with some hypothetical data to illuminate the concept of dispersion and how it relates to our example of education.

Measures of dispersion A group of statistics that describe how tightly or loosely cases in a distribution are clustered.

An Example of Two Communities. Imagine two counties, each with a population with an average of 12 years of education. In the first county, every citizen has 12 years of education; but in the second county half the population has 0 years of education and the other half has 24 years of education. Analyzing only the average may lead a researcher to believe that these are very similar counties; however, upon closer examination of how the average of 12 years is divided among the population, one would conclude that they are nothing alike. Without measures of dispersion, we know nothing about how widely education is distributed around the average years of education for a population.

Now suppose we have two groups of students, Group A and Group B, with six in each group ($N = 6$). Suppose Group A graduated from one college and Group B graduated from a different college. Five years after graduation, each group averages $50,000 in annual income. Therefore, for each group the mean is equal to $50,000.

TABLE 3 *Education by Race*

RACE OF RESPONDENT			FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
WHITE	Valid	Less than 12 Years	428	13.0	13.0	13.0
		12 Years	917	27.9	27.9	41.0
		More than 12 Years	1939	59.0	59.0	100.0
		Total	3284	100.0	100.0	
BLACK	Valid	Less than 12 Years	137	21.6	21.6	21.6
		12 Years	165	26.0	26.0	47.6
		More than 12 Years	332	52.4	52.4	100.0
		Total	634	100.0	100.0	
OTHER	Valid	Less than 12 Years	216	36.5	36.5	36.5
		12 Years	122	20.6	20.6	57.1
		More than 12 Years	254	42.9	42.9	100.0
		Total	592	100.0	100.0	

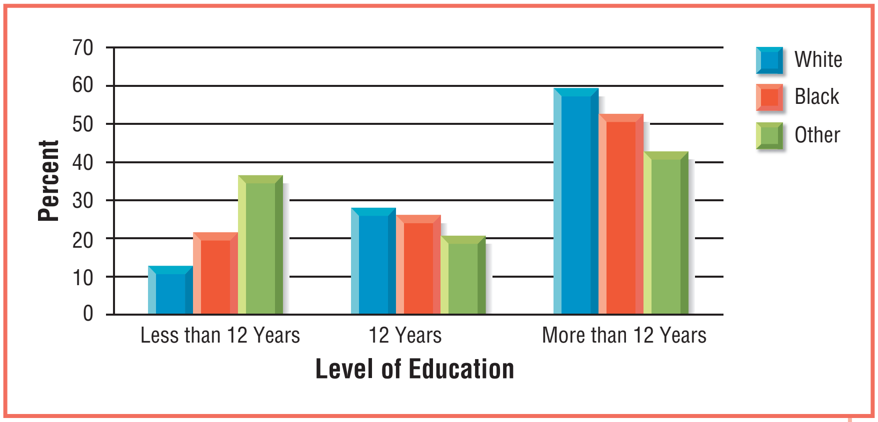


Figure 1 **Distribution of Education by Race**

This does not mean, however, that every person in each group is making $50,000, as shown in Figure 2.

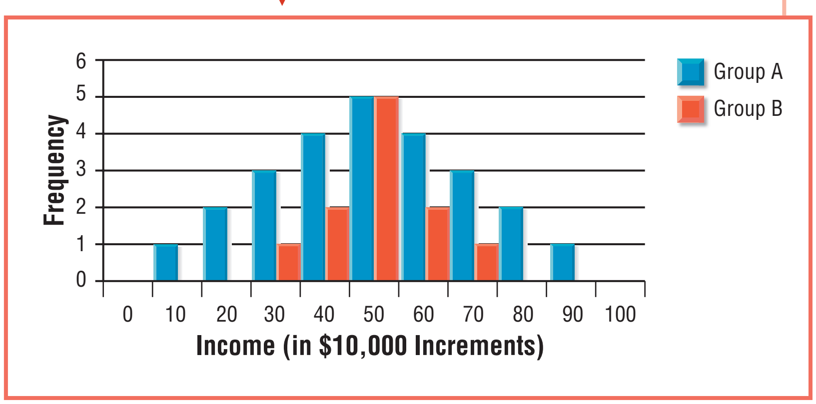
Despite having the same mean, it is clear that these are two very different distributions with Group B being more tightly clustered around the mean than Group A. We can begin to describe this difference statistically using a histogram.

As noted at the beginning of this chapter, data from the 2006 General Social Survey indicate that Americans, on average, have about 13.29 years of education. Yet this is not true for all groups. For Whites, the mean is 13.66; for Blacks, the mean is 12.87; and for Others the mean is only 11.69, two years less than it is for Whites. Histograms generated in SPSS for Windows show this difference in Figures 3, 4, and 5.

Three statistics are presented in the upper right-hand corner of each chart: the mean, standard deviation, and number of cases. The groups are similar in that each contains respondents that have between 0 and 20 years of education; however, they are not identical distributions. You can see that the cases in the third chart (Others) are more spread out than they are for the first two charts. This is particularly true for the height of the bar at the lower end of the distribution, respondents with less than 8 years of education. This indicates a greater degree of dispersion among Others relative to the degree of dispersion among Whites and Blacks. This is confirmed by the value of the second statistic in each chart, the standard deviation (*s*).

As you can see, the standard deviation in years of education for Whites is 2.929 and for Blacks is 2.951. These are very similar values. For Others, however, the standard deviation is equal to 4.360. It is this significantly higher standard deviation that makes

Figure 2 **Histograms of Income for Group A and Group B**



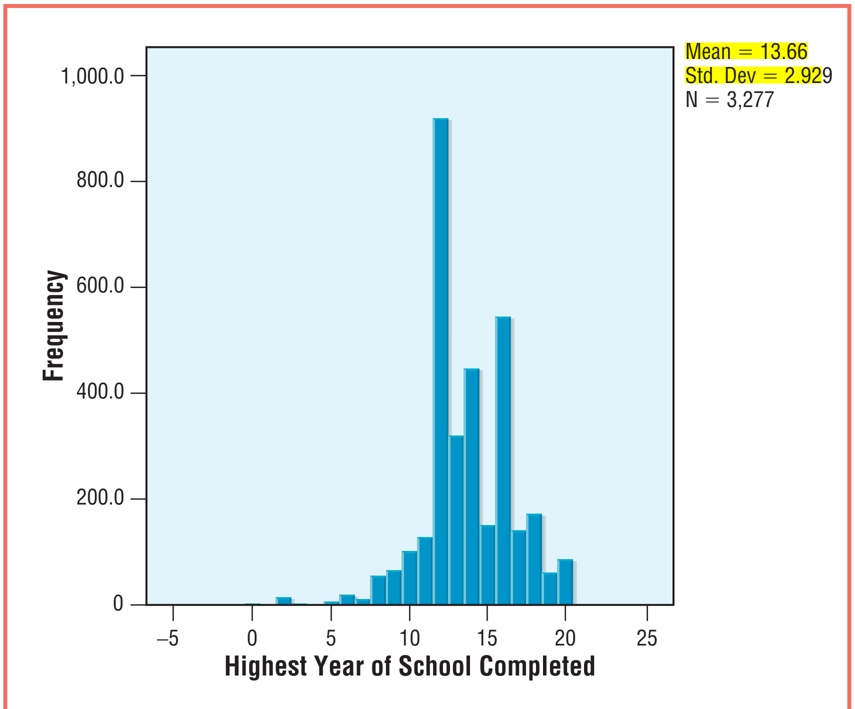
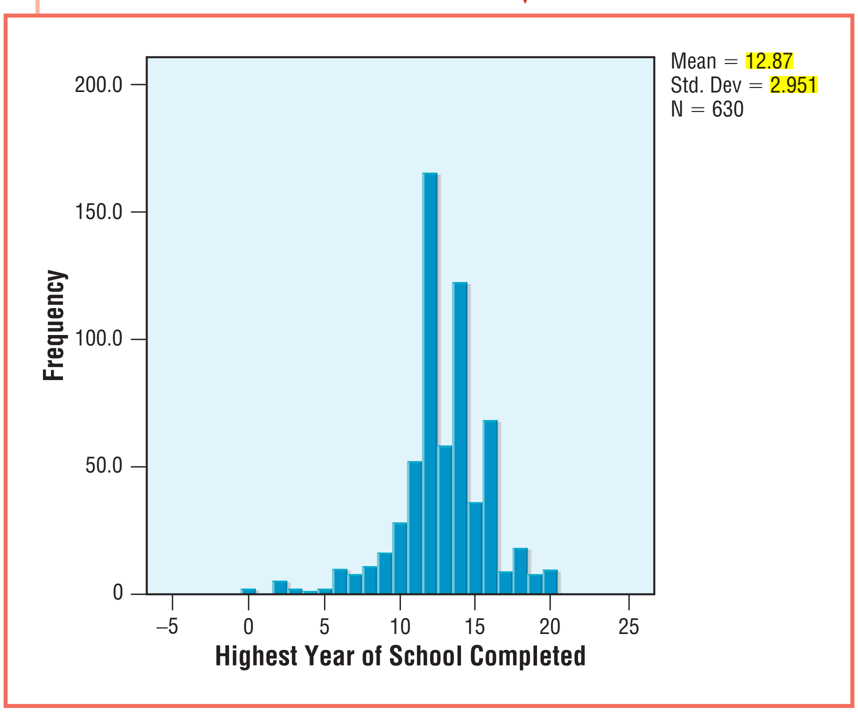


Figure 3 **Race of Respondent: White**

Figure 4 **Race of Respondent: Black**



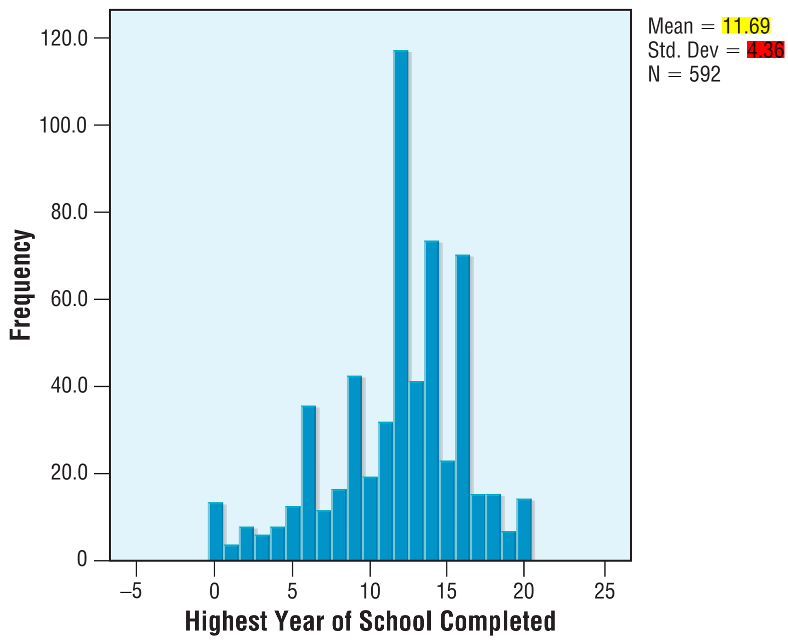


Figure 5 **Race of Respondent: Other**

the histogram for Others look different. The larger the standard deviation, the more widely distributed are cases around the mean. But let's not get ahead of ourselves! It is important to begin this discussion with an investigation into some other measures of dispersion before we get to the standard deviation. The range, interquartile range, mean deviation, variance, and standard deviation all help us to describe this difference. We now turn our attention to these five measures of dispersion, beginning with the range.

Range

The range is found by subtracting the lowest value in a set of data from the highest. The range offers a rough idea of how much variation there is in the categories of a variable.

Range The difference between the values of the highest and lowest cases in a distribution.

Using the earlier example of college students in Groups A and B in Figure 2, the highest value in Group A is $90,000 and the lowest value is $10,000. Using the formula for the range, we see that:

$$R = H - L$$
$$R = 90{,}000 - 10{,}000 = 80{,}000$$

The range is equal to $80,000 for Group A. For group B, we see that:

$$R = 70{,}000 - 30{,}000 = 40{,}000$$

The range is equal to $40,000 for Group B. Therefore, the respondents in Group A are more widely dispersed than are the respondents in Group B.

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The range can also apply to ecological data. For example, we may want to know more about the distribution of waste sites in communities across the state. The Commonwealth of Massachusetts is home to 351 minor civil divisions (towns and cities) and over 30,000 hazardous sites. These sites, however, are not evenly distributed across the 351 minor civil divisions. The city of Boston is the community with the most ecological hazards (3,972 points) whereas other communities, such as Leyden, have 0 points. Therefore, the range for this data is equal to

$$R = H - L$$
$$R = 3972 - 0 = 3972$$

Overall, the range is a rather simplistic measure of dispersion that requires us to use only two cases in a distribution to determine its value, the one with the highest value and the one with the lowest value. In this way, it provides us with a rough idea of the degree of dispersion, but by virtue of omitting the values of all but two of the cases, a great deal of data is never put to use. Other measures of dispersion, such as the mean deviation, variance, and standard deviation, use more data and offer a more complete understanding of our data by using the values of more cases in their calculations.

Integrating Technology

Although the range is relatively easy to calculate, it is often difficult to determine the highest and lowest values when working with very large data sets. Both SPSS and Excel have a "Sort Data" function that orders cases in either ascending or descending values of any variable the user selects. Then, by looking at the values of the first and last cases, the range can be easily calculated. The two tables below represent identical data on the Socioeconomic Index (SEI) taken from the 2006 General Social Survey. In the first, data are unsorted and in the second they are sorted from lowest to highest.

UNSORTED DATA	SORTED DATA
35.7	28.4
32.0	28.6
43.7	29.2
34.4	32.0
36.5	32.3
36.5	34.4

UNSORTED DATA	SORTED DATA
78.5	35.7
32.3	36.5
38.4	36.5
62.0	38.4
44.7	41.6
49.6	43.7
63.5	44.7
45.3	45.3
29.2	49.6
28.4	50.7
41.6	62.0
68.8	63.5
68.1	68.1
91.2	68.1
76.5	68.8
50.7	76.5
68.1	78.5
28.6	91.2

Once data have been sorted, it is much easier to calculate the range:

$$R = H - L$$

Because the range only uses two of the 24 values of data, a great deal of information is left out. As we will see, the mean deviation, variance, and standard deviation make use of all available data. Although they are more difficult to calculate, they give us a better overall picture of the distribution.

Interquartile Range

Another type of range is called the interquartile range. As you may have guessed, the interquartile range is based on dividing a distribution into quarters, four equal-sized parts known as quartiles. The difference between the value of the cases at the third and first quartiles is equal to the interquartile range. Literally, it is the range of values of the two middle quartiles.

Interquartile range The difference between the values at the third and first quartiles in a distribution.

For example, suppose that we sample 20 individuals on the number of years of school they have completed and get the following results (ranked in order from lowest to highest).

8 10 10 10 12 12 12 12 12 12 12 14 14 16 16 16 16 17 18 20

The median is equal to the value of the middle case. The formula for finding the middle case is $\frac{N+1}{2}$ or $(N + 1)(.5)$. Because two quartiles is equal to one half, the second quartile is the same thing as the median. Therefore, the data above the second quartile is equal to,

$$Q_2 = (20 + 1)(.5) = \textit{Value of case number } 10.5 = 12$$

When we count over 10.5 cases, we find that it falls between two values of 12. Therefore, $Q_2 = 12$, as shown below.

25% 25% 25% 25%

8 10 10 10 12 | 12 12 12 12 12 | 12 14 14 16 16 | 16 16 17 18 20

Low Score Q_1 Q_2 (median) Q_3 High Score

We use a similar method to find the location of the first and third quartiles. For the first quartile, we use the formula $N + 1(.25)$ and for the third quartile we use the formula $N + 1(.75)$. Therefore, in our example we find that,

$$Q_1 = \textit{Value of case number } (20 + 1)(.25) = \textit{Value of case number } 5.25 = 12$$

and

$$Q_2 = \textit{Value of case number } (20 + 1)(.75) = \textit{Value of case number } 15.75 = 16$$

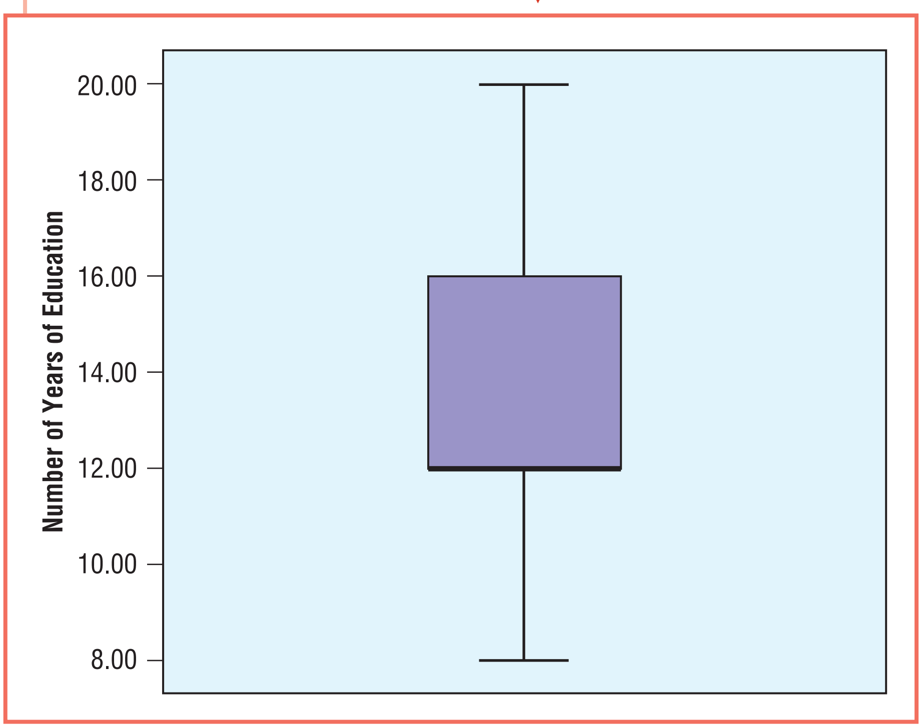
We can now calculate the interquartile range using the formula,

$$Q = Q_3 - Q_1 = 16 - 12 = 4$$

Box Plot A way to graphically demonstrate a distribution using the median, the 25th percentile, the 75th percentile, and the high and low values in a data set.

The interquartile range gives us a sense of the amount of dispersion across the middle part of a distribution. It does not, however, tell us much about dispersion that may be located in the tails of a distribution. SPSS for Windows can quickly create what are called *box plots*, charts like the one shown in Figure 6. You can see that the

Figure 6 **Box Plot of Years of Education**



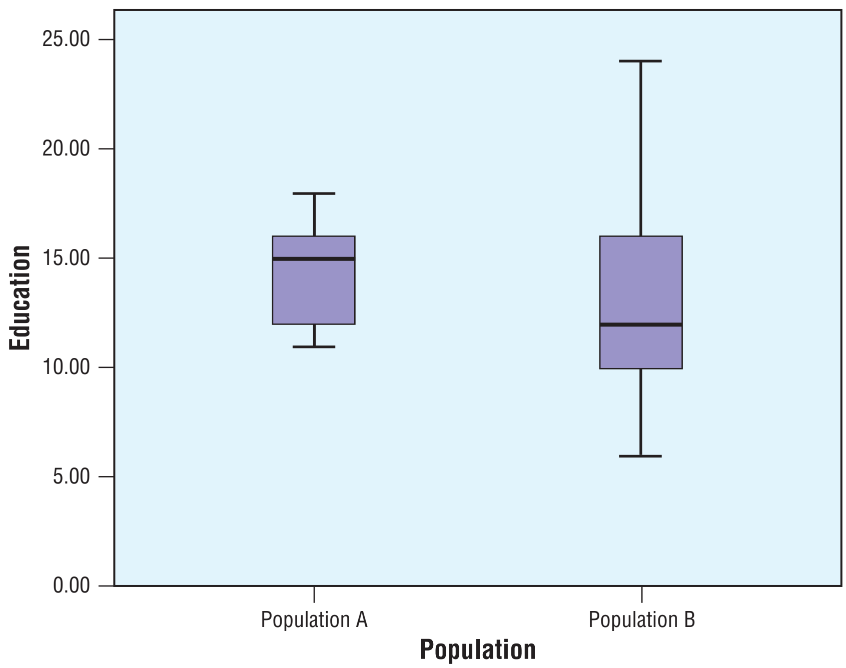


Figure 7 **Box Plots for Two Populations, A and B**

box plot contains a box in the center that ranges from values of 12 to 16. This box represents the interquartile range of our data. The chart also has lines that indicate the highest and lowest values of our data. The horizontal line, at the bottom of the box in this example, represents the median. Box plots are a good way to quickly assess the characteristics of a distribution and a good way to make comparisons between different groups. For example, use the box plots in Figures 6 and 7 to compare years of education between two populations.

As you can see from Figure 7, Population A has much less overall dispersion and a much smaller interquartile range than does Population B. The median is also significantly higher for Population A. Overall, while some people in Population B have very high levels of education (and very low levels of education) that far exceed anyone in Population A, as a group, Population A is more highly educated.

Mean Deviation

The mean deviation, as you might have guessed from its name, is a type of average. Although it is rarely used, a brief summary is offered here as a way to introduce concepts that are useful in understanding a much more commonly used measure of dispersion, the standard deviation. The mean deviation differs from the mean because it is based on the degree to which cases differ from the mean rather than the actual values of the cases. For example, if the average number of courses that students take is five and you

Mean deviation The mathematical average of deviations from the mean (measured in absolute values).

take four, you vary from the mean by a value of 1. This amount is called the deviation and can be calculated by taking the absolute value of the difference between the value of a particular case and the mean, using the following formula:

$$D_i = |X - \overline{X}|$$

When calculating the mean deviation, this is done for each case in the distribution. The values are then summed and divided by N to obtain the mean deviation.

$$D_t = \frac{\Sigma(|X - \overline{X}|)}{N}$$

It is necessary to use the absolute values of the deviations from the mean, otherwise, the values will add up to 0. Because absolute values are used, the mean deviation can also be called the mean absolute deviation. This mathematical certainty is a unique characteristic of the mean.

Mean Deviation Using Raw Data

If we take any group of numbers, suppose it is the number of courses that students at your school take, the average represents the point at which half of the total deviation falls above and half falls below. Therefore, if we sum the positive and negative deviations they will add to 0.

Consider this example: Suppose we ask six students how many courses they are currently enrolled in and we get the results shown below.

4
5
4
6
5
6

The values add to 30 and because $N = 6$, the mean is 5. If we subtract the mean from each value, we find:

$$\begin{aligned}
4 - 5 &= -1 \\
5 - 5 &= 0 \\
4 - 5 &= -1 \\
6 - 5 &= 1 \\
5 - 5 &= 0 \\
6 - 5 &= 1 \\
\hline
\text{Sum} &= 0
\end{aligned}$$

This is the case for any group of numbers and requires us to use absolute values when calculating the mean deviation. When variations from the mean are put in absolute value (positive value), we find:

$$\begin{aligned}
4 - 5 &= -1 = 1\\
5 - 5 &= 0 = 0\\
4 - 5 &= -1 = 1\\
6 - 5 &= 1 = 1\\
5 - 5 &= 0 = 0\\
6 - 5 &= 1 = 1\\
\hline
\text{Sum} &= 4
\end{aligned}$$

Using the formula for the mean deviation, we find:

$$D_t = \frac{\Sigma(|X - \overline{X}|)}{N} = \frac{4}{6} = .67$$

We can now say that the mean deviation is equal to .67. This means that, *on average*, cases deviate from the mean by .67. On average, students deviate from the mean number of courses by .67 courses.

Now You Try It 1

Use the two data sets below to calculate the range and mean deviation. In what way are the two groups similar? In what ways are they different?

Number of Class Meetings Missed in the Past Semester

Males	Females
3	7
7	4
9	9
12	8
1	6
8	8
2	5
14	9

Hint #1: To calculate the range, begin by rank-ordering the cases from lowest to highest.

Hint #2: When finding the mean deviation, remember to first calculate the mean. Then determine how much each case differs from the mean in absolute value.

Variance and Standard Deviation

We are now ready to get back to our discussion of the standard deviation. While the mean deviation gives us an idea about the nature of dispersion in our data, it is not a widely used measure. Also based on the concept of deviations from the mean are two more commonly used measures, the variance and the standard deviation. These are practically the same measure. In fact, they are identical except that one is expressed in squared units while the other is expressed in standard units. To obtain the standard deviation (the most commonly used measure of dispersion), we need only to take the square root of the variance. Therefore, we now turn our attention first to the variance and then to the standard deviation.

Variance The mathematical average of the squared deviations from the mean.

Blend Images/Alamy

Variance (s^2)

Like the mean deviation, the variance is an average based on deviations from the mean. And like the calculations for the mean deviation, it is necessary to make sure that the deviations from the mean are expressed as positive values. Otherwise, the sum of deviations from the mean will add up to 0. Rather than taking the absolute values of deviations from the mean, the variance squares the deviations from the mean. While this has the effect of turning all of these values into positive numbers, it also has the effect of altering the scale of their values. In other words, 3s become 9s, 9s become 81s, and so on.

In this sense, it is important to remember that the variance, (s^2), is always expressed in "squared units," while the standard deviation, (s), is expressed in standard units. The variance is determined by summing the squared deviations from the mean and dividing by N.

The formula for the variance is:

$$s^2 = \frac{\Sigma(X - \overline{X})^2}{N}$$

We now calculate the variance using data on television viewing hours per day for two groups: Group A and Group B.

EXAMPLE 1

Calculating the Variance Using Raw Data

Our data for Group A is:

3, 2, 4, 2, 4

Using the same technique that we used to determine deviations from the mean, we subtract the mean from each value ($X - \overline{X}$) in Table 4. The sum of these squared deviations from the mean is equal to the *sum of squares* and is denoted by the symbol *ss*.

TABLE 4

X	$X - \overline{X}$	$(X - \overline{X})^2$
3	$3 - 3 = 0$	0
2	$2 - 3 = -1$	1
4	$4 - 3 = 1$	1
2	$2 - 3 = -1$	1
4	$4 - 3 = 1$	1
		ss = 4

Because the deviations from the mean are all equal to either 0 or 1, the squared deviations from the mean are also equal to either 0 or 1. This means that for this particular example, the sum of squares is the same as the sum of absolute value deviations, but only because we are working with only 0s and 1s.

We can now put the sum of squares (the numerator) and N (the denominator) into the equation for the variance.

$$S^2 = \frac{\Sigma(X - \overline{X})^2}{N} = \frac{4}{5} = .8$$

So in this example the variance is equal to .8. We now calculate the variance for Group B.

For Group B the data is:

1, 0, 7, 6, 1

Using the same technique that we used to determine deviations from the mean, we subtract the mean from each value $(X - \overline{X})$ in Table 5. The sum of these squared deviations from the mean is equal to the *sum of squares* and is denoted by the symbol *ss*.

TABLE 5

X	$X - \overline{X}$	$(X - \overline{X})^2$
1	$1 - 3 = -2$	4
0	$0 - 3 = -3$	9
7	$7 - 3 = 4$	16
6	$6 - 3 = 3$	9
1	$1 - 3 = -2$	4
		$ss = 42$

We can now put the sum of squares and N into the equation for the variance.

$$s^2 = \frac{\Sigma(X - \overline{X})^2}{N} = \frac{42}{5} = 8.4$$

As you can see, 8.4 is an unusually large number, particularly since our data values only go up to a maximum of 7. This is a result of squaring the difference between the case values and the mean. In other words, our variance of 8.4 is expressed in a different unit of analysis than that of the data. To make the results more intelligible, we must remove the squaring effect by taking the square root of the variance. By doing so, we have calculated the standard deviation.

Standard Deviation (s)

Standard deviation The square root of the mathematical average of the sum of squared deviations from the mean. The square root of the variance.

The standard deviation is equal to the square root of the variance. Therefore, to calculate the standard deviation, it is necessary to first calculate the variance. Once we have calculated the variance, we simply take the square root and the result is the standard deviation.

EXAMPLE 2

Calculating the Standard Deviation Using the Variance

Therefore, for Group A, the variance (s^2) is equal to .8. By taking the square root of .8, we find that:

$$s = \sqrt{s^2} = \sqrt{.8} = .9$$

The standard deviation is equal to .9. This means that the cases differ from the mean of 3 hours of television viewing by an average of .9 hours (56 minutes).

For Group B, the variance (s^2) is equal to 8.4. By taking the square root of 8.4, we find that:

$$s = \sqrt{s^2} = \sqrt{8.4} = 2.9$$

The standard deviation is equal to 2.9. This means that each case differs from the mean of 3 hours of television viewing by an average of 2.9 hours (174 minutes).

It is important to remember that the variance and standard deviation are averages. They are averages of how much cases tend to differ from the mean. The variance is expressed in units of analysis that have been squared and the standard deviation is expressed in standard units (meaning nonsquared units) of analysis (hence the name *standard* deviation).

EXAMPLE 3

Calculating the Standard Deviation Using Raw Data

Suppose we are interested in learning more about intergenerational differences in education. Specifically, we want to know how much more or less formal education people have compared to their grandparents. To investigate, we gather data from 10 females, each of whom is 40 years old. First, we ask them how many years of formal education they have. Then we ask them how many years of formal education their grandmother had when she was 40 years old. The difference between the two is calculated as follows:

Respondent's Education – Respondent's Grandmother's Education $=$ Change in Education

The result can be either positive (for cases in which the respondent has more years of education than did her grandmother) or negative (for cases in which the respondent has less years of education than did her grandmother). In this example, all respondents have as much or more education than did their grandmothers at the same age. Our data is presented below in Table 6.

TABLE 6

RESPONDENT #	RESPONDENT'S EDUCATION	RESPONDENT'S GRANDMOTHER'S EDUCATION	CHANGE IN EDUCATION
1	12	10	2
2	16	12	4
3	12	10	2
4	14	10	4
5	14	12	2
6	12	12	0
7	12	8	4
8	16	12	4
9	16	14	2
10	12	10	2

To begin, we calculate the means as follows:

$$\bar{X} = \frac{\Sigma X}{N} = \frac{2 + 4 + 2 + 4 + 2 + 0 + 4 + 4 + 2 + 2}{10} = \frac{26}{10} = 2.6$$

We now determine how much each case differs from the mean in Table 7.

TABLE 7

CHANGE IN EDUCATION (X)	DEVIATION FROM THE MEAN ($X - \bar{X}$)
2	$2 - 2.6 = -.6$
4	$4 - 2.6 = 1.4$
2	$2 - 2.6 = -.6$
4	$4 - 2.6 = 1.4$
2	$2 - 2.6 = -.6$
0	$0 - 2.6 = -2.6$
4	$4 - 2.6 = 1.4$
4	$4 - 2.6 = 1.4$
2	$2 - 2.6 = -.6$
2	$2 - 2.6 = -.6$

These values are then squared and summed to give us the *sum of squares* in Table 8.

TABLE 8

DEVIATION FROM THE MEAN $(X - \bar{X})$	SQUARED DEVIATIONS FROM THE MEAN
−.6	.36
1.4	1.96
−.6	.36
1.4	1.96
−.6	.36
2.6	6.76
1.4	1.96
1.4	1.96
−.6	.36
−.6	.36
	Sum of Squares = 16.4

The next step is to obtain the variance by dividing the sum of squares by the number of cases. Therefore,

$$S^2 = \frac{SS}{N} = \frac{16.4}{10} = 1.6$$

The final step in our equation is to determine the standard deviation by taking the square root of the variance. Therefore,

$$S = \sqrt{S^2} = \sqrt{1.6} = 1.26$$

This tells us that the average deviation from the mean is 1.26 years of education. At first, this may appear to be a small value, but remember that we are looking at the average deviation in changes in intergenerational education. The average change in intergenerational education is 2.6 years with a standard deviation of 1.26 years.

A Note on Samples and Populations (*N* or *N* – 1?). You may have noticed that if you input these same data into SPSS for Windows, you will find that you do not get the same results. Instead, you get a variance of 1.82 and a standard deviation of 1.35. The reason for this is that the SPSS for Windows software does not use N in its calculation of these statistics. Instead, the variance and standard deviation are based on $N - 1$. The reason for this is that sample data and population data are treated differently. Up to this point, we have only discussed statistics in ways that assume that our data are perfectly reflective of the population from which they were drawn. Because this is seldom, if ever the case, statisticians have devised a set of tools to control for the differences between sample data and population data. One such tool is to use $N - 1$ in place of N.

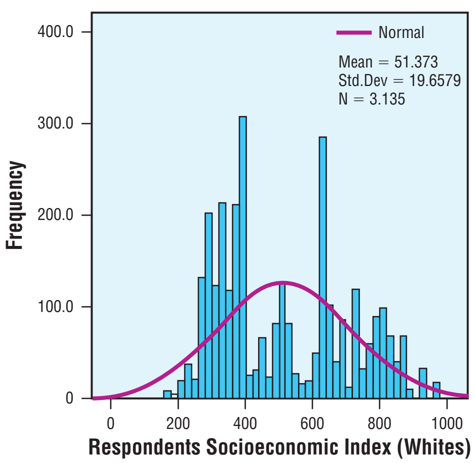
At this point, you might want to check with your instructor to see if he or she wants you to use N or $N - 1$ in the denominator of these equations. Different instructors prefer different methods. It is also important to realize that the difference between using N and $N - 1$ diminishes as sample size increases and that significant differences are only seen when working with a small number of cases. The examples and practice problems in this chapter are based on using N in the calculations.

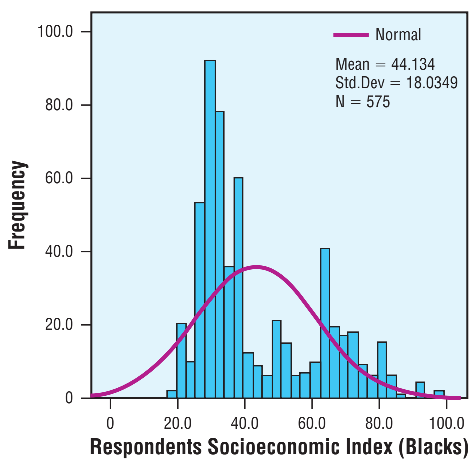
We now turn our attention to calculating measures of dispersion with grouped data.

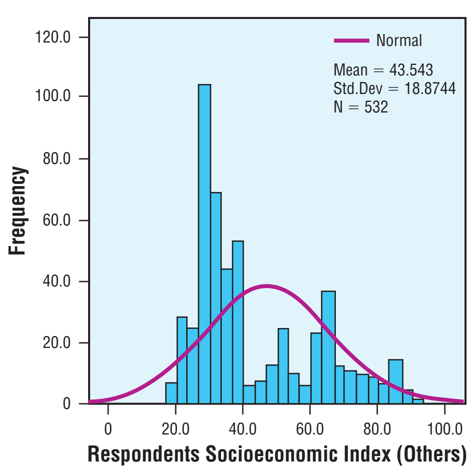
Integrating Technology

A variety of data analysis programs offer users the ability to work with data quickly and easily. Histograms and normal curves offer visualizations of how cases are distributed. SPSS for Windows provides users with the ability to quickly generate histograms with normal curve distributions of data. For example, suppose we have two groups of respondents (one black and one white) and we want to compare their overall socioeconomic standing (in this case a measure based on occupation and prestige). SPSS for Windows allows us to quickly create different groups by "splitting" the database by race. We can then create a histogram with a normal curve for each group as shown below.

As you can see, the mean SEI score is 51.4 among Whites, 44.1 among Blacks, and 43.5 among Others. The difference between the bars and the normal curve is that the







bars are our actual observations while the curve represents our predicted observations. As you can see, each group has a significant number of cases with low values, relatively few cases with "average" values, and slightly more cases than expected with high values. The difference between the bars and the curve indicate that SEI is not normally distributed. Identifying this discrepancy in the charts is relatively easy; however, without graphically presenting data, it is usually only possible to the very well-trained statistician.

The normal curve and its unique properties are discussed in greater detail later in this chapter.

EXAMPLE 4

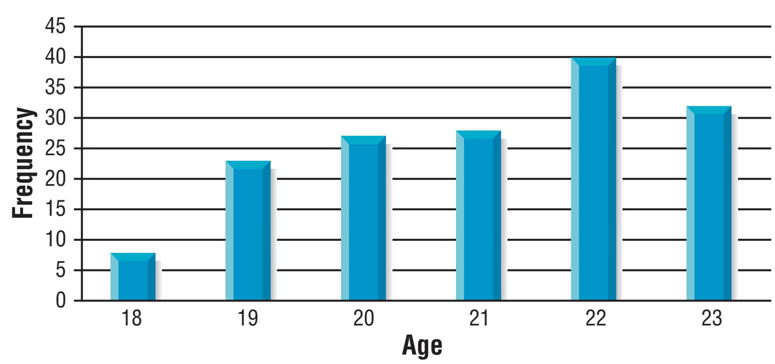
Calculating Variance and Standard Deviation Using Grouped Data

Suppose we sample a group of students living in an off-campus housing facility to better understand the distribution of age. Our data yield the following frequency table in Table 9 and histogram in Figure 8.

TABLE 9 *Age of Respondent*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	18	8	5.1	5.1	5.1
	19	23	14.6	14.6	19.6
	20	27	17.1	17.1	36.7
	21	28	17.7	17.7	54.4
	22	40	25.3	25.3	79.7
	23	32	20.3	20.3	100.0
	Total	158	100.0	100.0	

Figure 8 **Age of Respondent**



The mean age of the students is 21 years ($\bar{X} = 21$). As you can see from the first row (18-year-olds) in Table 10, our process for calculating the variance and standard deviation is essentially the same as when we work with raw data. To find the sum of squares using grouped data:

1. Find the difference between an 18-year-old's age and the mean.
2. Square that amount. At this point, we have found the squared difference between only one 18-year-old and the mean; but we have eight 18-year-olds.
3. Multiply the squared difference from the mean by the frequency for that age group. In this case we multiply 9 (the squared difference) by 8 (the frequency of 18-year-olds) to get 72.
4. Repeat steps 1–3 for each age group and add the results to get the sum of squares.
5. Divide the sum of squares by N to get the variance.
6. Take the square root of the variance to get the standard deviation.

To calculate the variance:

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{N} = \frac{359}{158} = 2.3$$

To calculate the standard deviation:

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}} = \sqrt{\frac{359}{158}} = \sqrt{2.3} = 1.5$$

This tells us that while the mean age of the group of students is 21, the average deviation from the mean is 1.5 years.

TABLE 10

X	$X - \bar{X}$	$(X - \bar{X})^2$	$(X - \bar{X})^2 f$
18 ($f = 8$)	−3	9	9 * 8 = 72
19 ($f = 23$)	−2	4	4 * 23 = 92
20 ($f = 27$)	−1	1	1 * 27 = 27
21 ($f = 28$)	0	0	0 * 28 = 0
22 ($f = 40$)	1	1	1 * 40 = 40
23 ($f = 32$)	2	4	4 * 32 = 128
			Sum of Squares = 359

Now You Try It 2

Use the data below to calculate the mean, variance, and standard deviation. (*Hint:* You need to calculate the mean first.)

9 7 6 5 1 3 13 6 8 9 10

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{N}$$

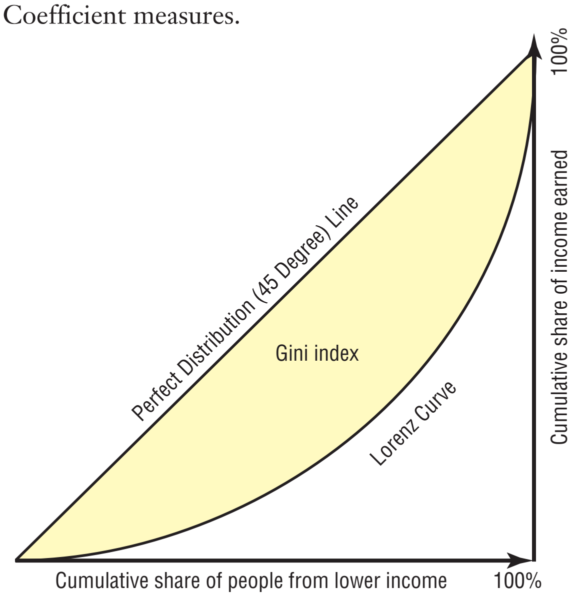
$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N}}$$

Statistical Uses and Misuses

Other Ways of Thinking about Dispersion: The GINI Coefficient

Corrado Gini (1864–1965) was an Italian statistician who wanted to find a way to measure the overall level of income inequality in society. Once this was accomplished, it would then be possible to apply the measure to different societies as a way to compare one country to another. Ultimately, he was able to develop a measure that was later named after him, the GINI Coefficient.

The GINI Coefficient is a number that ranges between 0 and 1.0. While it can be used to describe the overall distribution of virtually any variable, it is very often used to describe the overall distribution of income in a town, county, region, or country. The chart below helps us to interpret what the GINI Coefficient measures.



(Daniel J. Evans School of Public Affairs, 2009)

As you can see, the *x*-axis represents the percent of the population and the *y*-axis represents the percent of income. A perfectly even distribution of income would mean that 10% of the population received 10% of the income, 20% of the population received 20% of the income, and so on. This results in a straight line. Should 10% of the population receive only 1% of the income, the line begins to sag. The more sag in the line, the greater the degree of inequality of income distribution.

The actual GINI Coefficient is the proportion of area under the triangle that is between the sagging line and the even line.

The United Nations *Human Development Report* lists the GINI Coefficient for income for practically all of the nations of the world. It is interesting to note that many Americans may tend to believe that the GINI Coefficient in the United States (45.0) would be similar to western European nations such as the United Kingdom (34.0), France (28.0), and Germany (28.0); however, the nations with similar GINI Coeffients are actually Kenya (44.5), Philippines (44.5), Cameroon (44.6), Cote d'Ivoire (44.6), Uruguay (45.2), Jamaica (45.5), Uganda (45.7), and Ecuador (46.0).

Remember that this does not mean the United States has similar income levels. Income in the U.S. is much higher than these countries. The way that income is distributed, however, is similar to these countries. In other words, when measured using the GINI Coefficient, income distribution in the United States represents that of developing nations (sometimes referred to as third world nations).

Now let's take a look at some toxic waste data from the Massachusetts study (Faber & Krieg, 2005) using towns that have somewhere between 25 and 50 waste sites. A frequency table of the results is shown in Table 11. The mean number of sites is 37.

We can calculate the sum of squares the same way we would if working with raw data. The only difference is that we multiply the squared difference between X and the mean by the number of times X occurs. In other words, we multiply it by the frequency of X. This is shown in Table 12.

Now that we have determined that the sum of squares is equal 4,485, we can divide it by N ($N = 76$) to get the variance.

$$s^2 = \frac{\Sigma(X - \overline{X})^2}{N} = \frac{4485}{76} = 59.0$$

TABLE 11 *Frequency of Sites*

SITES		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	25	5	6.6	6.6	6.6
	26	3	3.9	3.9	10.5
	27	5	6.6	6.6	17.1
	28	1	1.3	1.3	18.4
	29	4	5.3	5.3	23.7
	30	2	2.6	2.6	26.3
	31	2	2.6	2.6	28.9
	32	1	1.3	1.3	30.3
	33	3	3.9	3.9	34.2
	34	2	2.6	2.6	36.8
	35	5	6.6	6.6	43.4
	36	1	1.3	1.3	44.7
	38	4	5.3	5.3	50.0
	39	2	2.6	2.6	52.6
	40	5	6.6	6.6	59.2
	41	4	5.3	5.3	64.5
	42	6	7.9	7.9	72.4
	43	2	2.6	2.6	75.0
	44	3	3.9	3.9	78.9
	45	4	5.3	5.3	84.2
	46	2	2.6	2.6	86.8
	47	1	1.3	1.3	88.2
	48	2	2.6	2.6	90.8
	49	6	7.9	7.9	98.7
	50	1	1.3	1.3	100.0
	Total	76	100.0	100.0	

By taking the square root of the variance, we get the standard deviation.

$$s = \sqrt{\frac{\Sigma(X - \overline{X})^2}{N}} = \sqrt{\frac{4485}{76}} = \sqrt{59.0} = 7.7$$

This tells us that, while the mean number of waste sites in these towns is 37, towns typically have within 7.7 sites of the mean. This is useful information, but it also begs the question of just what we mean by "typical." The standard deviation and the normal curve share some interesting properties that tell us just what is meant by "typical." We now shift our discussion to why the standard deviation is such an important statistical tool.

TABLE 12

X	f	$X - \bar{X}$	$(X - \bar{X})^2$	$(X - \bar{X})^2 f$
25	5	25 − 37 = −12	144	144 * 5 = 720
26	3	26 − 37 = −11	121	121 * 3 = 363
27	5	27 − 37 = −10	100	100 * 5 = 500
28	1	28 − 37 = −9	81	81 * 1 = 81
29	4	29 − 37 = −8	64	64 * 4 = 256
30	2	30 − 37 = −7	49	49 * 2 = 98
31	2	31 − 37 = −6	36	36 * 2 = 72
32	1	32 − 37 = −5	25	25 * 1 = 25
33	3	33 − 37 = −4	16	16 * 3 = 48
34	2	34 − 37 = −3	9	9 * 2 = 18
35	5	35 − 37 = −2	4	4 * 5 = 20
36	1	36 − 37 = −1	1	1 * 1 = 1
38	4	38 − 37 = 1	1	1 * 4 = 4
39	2	39 − 37 = 2	4	4 * 2 = 8
40	5	40 − 37 = 3	9	9 * 5 = 45
41	4	41 − 37 = 4	16	16 * 4 = 64
42	6	42 − 37 = 5	25	25 * 6 = 150
43	2	43 − 37 = 6	36	36 * 2 = 72
44	3	44 − 37 = 7	49	49 * 3 = 147
45	4	45 − 37 = 8	64	64 * 4 = 256
46	2	46 − 37 = 9	81	81 * 2 = 162
47	1	47 − 37 = 10	100	100 * 1 = 100
48	2	48 − 37 = 11	121	121 * 2 = 242
49	6	49 − 37 = 12	144	144 * 6 = 864
50	1	50 − 37 = 13	169	169 * 1 = 169
	N = 76			Sum of Squares = 4485

Eye on the Applied

Public Education as Equalizer or Divider?

A great deal of debate takes place over the question of whether public education reduces overall inequality or reinforces existing patterns of inequality. Famous Brazilian educator Paulo Freire was one of the leading figures in *critical pedagogy* until his death in 1997. His most famous book is *Pedagogy of the Oppressed* (1970), Freire argued that by taking the view of students as "empty vessels" into which knowledge should be poured, schools fail to teach students how to challenge existing forms of domination and instead teach them to conform to existing power structures. Today, many educators and sociologists claim that Freire was correct and that standardized testing at the federal and state levels serves the interests of those in power rather than serving as a tool to bring about a more egalitarian society.

Increasingly, standardized testing is used to determine whether public schools are meeting the needs of their students. In New York, all students must pass basic-level math skills or forgo their high school diploma. Aside from the obvious problem of determining which skills are deemed important (there is no art or music exam), is the problem of how average test scores are used to assess school performance.

Many cities have what are called "magnate schools" (also called magnet schools) where students must meet certain criteria to gain admission. Such schools cater to "gifted and talented" students. These are public schools funded by local residents, yet not all children may attend. Not only must students test into these schools, they must maintain a grade point average (GPA) high enough to remain in the schools.

Generally, standardized testing results are made available at the end of the school year. In the case of Buffalo, New York, students unable to maintain a high enough GPA to remain in the magnate schools are transferred into nonmagnate schools before the standardized tests are given. Ultimately, this means that magnate schools are more likely to have those students most likely to pass, while nonmagnate schools take on the task of educating students that the magnate schools kicked out. This kind of process has the effect of making magnate schools appear to be stronger educational institutions, when, in fact, they inflate their average test scores by expelling weaker students into other schools.

We can therefore predict that the average test score will be higher in magnate schools and that the standard deviation of scores in magnate schools will be smaller than that of nonmagnate schools. Even if the schools have identical averages, it is likely that the variation around the average will be greater for nonmagnate schools.

If statistical analyses offer support for these ideas, it would raise serious questions about whether public education increases or decreases overall levels of inequality. What do you think about Freire's ideas?

Standard Deviation and the Normal Curve

As you probably know, normal curves (also known as bell curves) are evenly distributed. This means that if we were to split a normal curve down the center, each half would be a mirror image of the other half. An example of a normal curve is shown in Figure 9.

Normal curve A bell-shaped symmetrical distribution.

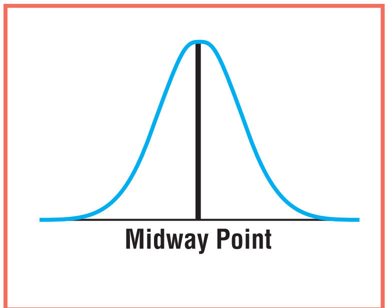


Figure 9 Normal Curve

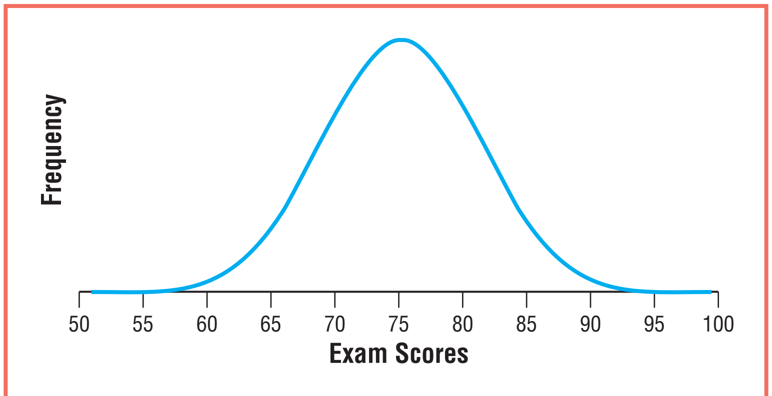


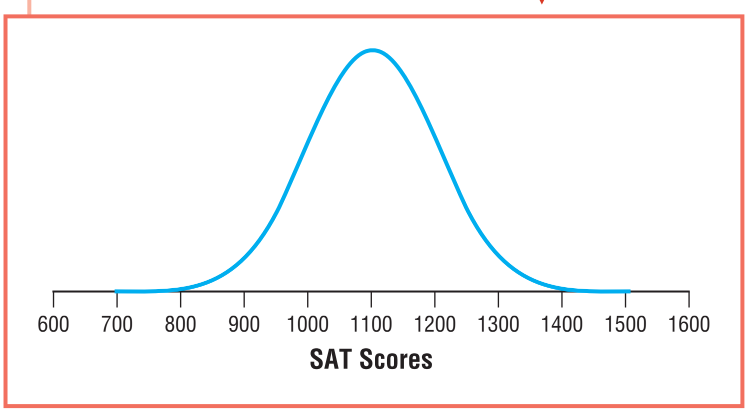
Figure 10 Distribution of Exam Scores

While a great many occurrences in the natural and social world tend to *approximate* normal curves (the age of students attending your school, the distribution of grade point averages, the amount of money you spend on books each semester, etc.), actual normal curves exist only as theoretical constructs. In other words, they exist only in theory, not in reality.

For example, suppose that a sociology professor gives students an exam. Because a C is considered average, we might expect the mean for the exam to be equal to a 75. Therefore, while very few students receive a 100, more students receive 90s, even more receive 80s, and the most common score would be a 75. Similarly, very few students would receive a 50, more would receive above a 50, even more receive 60s, and the most common score would be a 75. This is shown in Figure 10.

One of the unique properties of the normal curve is that all three measures of central tendency fall on the same value. In this example, the mean score is 75. This happens to be the point at which half the cases fall below and half above. Therefore, 75 is also the median exam score. And because 75 is the most frequently occurring exam score (the tallest point on the curve), it is also the mode.

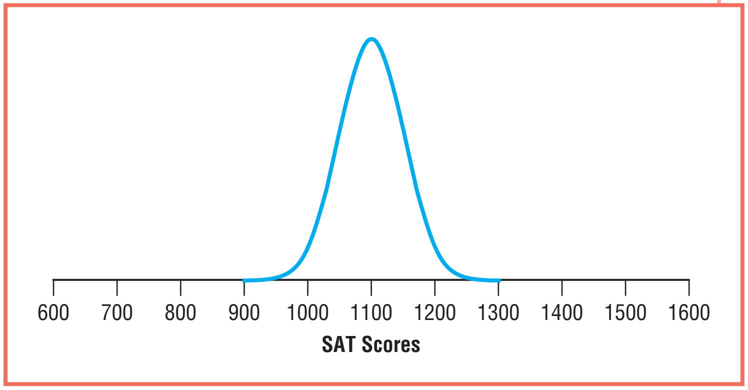
Figure 11 Group A: Distribution of SAT Scores (Mean = 1100, Std. Dev. = 100)



This brings us to the explanation of why measures of dispersion are so important when it comes to describing a distribution. It is possible to have two distributions with identical measures of central tendency (say 75), but different values for their standard deviation. For example, suppose we have two groups of students that each took the SAT. Each group averages a combined score of 1100; but while one group has a standard deviation of 100, the other has a standard deviation of 50. The scores are shown in Figures 11 and 12.

As you can see from the charts above, each group averaged 1100 on the SAT;

Figure 12 Group B: Distribution of SAT Scores (Mean = 1100, Std. Dev. = 50)

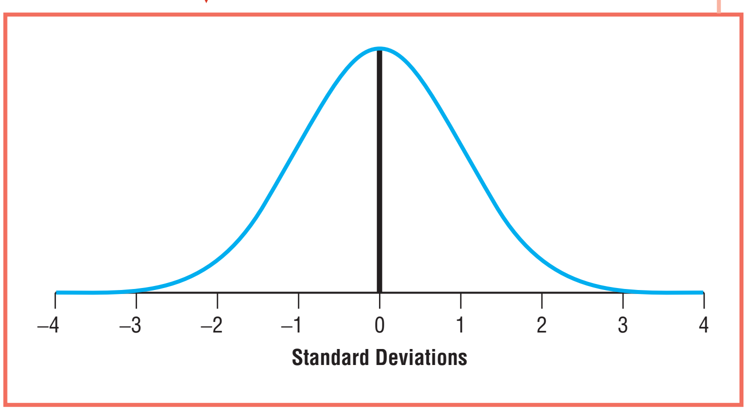


however, while one group is widely dispersed around the mean, the other group is more tightly clustered around the mean. As these charts demonstrate, it is necessary to have both measures of central tendency and measures of dispersion to more fully describe a distribution. While measures of central tendency tell us where cases tend to cluster (in this case around 1100), measures of dispersion tell us how spread out cases tend to be.

In addition to the highest point representing the mode, median, and mean, the normal curve has another important property. This time it is related to the standard deviation. For all normal curves, 68% of all cases fall within one standard deviation of the mean. This is true regardless of how clustered or how spread out the cases may be.

Using our examples of SAT scores, we see that $\overline{X} = 1100$ and $s = 100$. If 68% of all cases fall within one standard deviation of the mean, then 68% of all cases should fall within 100 points of 1100 points or 1100 ± 100. Therefore, for Group A, 68% of all

Figure 13 Normal Curve with Standard Deviations



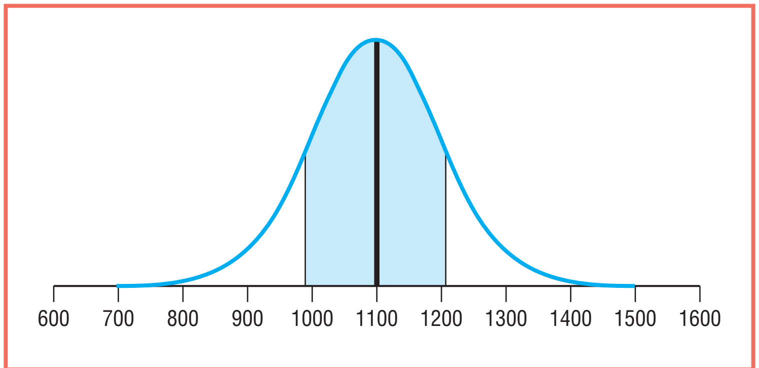


Figure 14 **Group A: Percentage of Cases that Fall within 1 Standard Deviation (±) of the Mean**

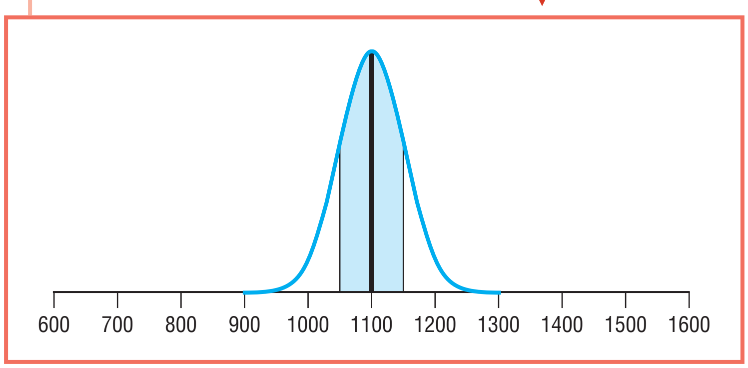
students received somewhere between 1000 and 1200 points on the SAT. When this is graphed it looks like Figure 14.

The shaded area to the left of the mean (1 standard deviation below the mean) constitutes roughly 34% of all cases and the shaded area to the right of the mean (1 standard deviation above the mean) constitutes roughly 34% of all cases. Together, the entire shaded area is equal to 68% of all cases. When we make a similar chart for Group B, it looks like Figure 15.

Each of the two charts contains a shaded area that goes from 1 standard deviation below the mean to one standard deviation above the mean. Each shaded area accounts for a total of 68% of the total area under the normal curve (68% of all cases). When we move to an additional standard deviation from the mean ($\pm 2s$), 95% of the total area under the curve (95% of cases) is included in the shaded area.

These unique properties of the normal curve are very important for the study of probability and make the standard deviation the most common of all measures of dispersion.

Figure 15 **Group B: Percentage of Cases that Fall within 1 Standard Deviation (±) of the Mean**



Now You Try It 3

Use the table below to calculate the variance and standard deviation. (*Hint:* The mean is equal to 2.9.)

Parking Tickets this Semester

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	1	2321	51.5	51.5	51.5
	2	440	9.8	9.8	61.3
	3	84	1.9	1.9	63.1
	4	143	3.2	3.2	66.3
	5	712	15.8	15.8	82.1
	6	139	3.1	3.1	85.2
	7	490	10.9	10.9	96.1
	8	177	3.9	3.9	100.0
	Total	4506	99.9	100.0	
Missing	NA	4	.1		
Total		4510	100.0		

Chapter Summary

This chapter focuses on measures of dispersion, statistics that are used to describe how cases in a distribution tend to be spread out. The three most commonly used measures of dispersion are the range, variance, and standard deviation. The variance and standard deviation are the most commonly used measures of dispersion and are interpreted in relation to the mean. The mean deviation is less common, but learning it plays an important role in understanding the overall concept of dispersion. All these measures are used only with interval/ratio measures.

While it is important to understand the differences between each of these, it is particularly important to understand how they differ from measures of central tendency. Remember, measures of central tendency tell what is typical about the values of cases in a distribution and measures of dispersion tell us how tightly cases tend to cluster in a distribution. Describing distributions using both measures of central tendency and measures of dispersion allows us to achieve a far more complete understanding of our data. Histograms play an important role by offering visual displays of distributions and the normal curve is very important because it places our understanding of the standard deviation in the context of the theory of probability.

Exercises

Use the data in the box below to calculate the mean, range, interquartile range, mean deviation, variance, and standard deviation. *Note:* For these exercises our calculations of the variance and standard deviation will use $N - 1$ in the denominator.

5	6	2	8	7	6	6	12	6	5	4	5

1. Mean: _____
2. Range: _____
3. Interquartile range: _____
4. Variance: _____
5. Standard deviation: _____

Find the range, interquartile range, variance, and standard deviation for the following sets of data:

Set A: 9, 6, 7, 4, 6, 4, 8, 7, 6, 3

6. Range: _____
7. Interquartile range: _____
8. Variance: _____
9. Standard deviation: _____

Set B: 9, 4, 18, 1, 16, 2, 17, 4, 15, 4

10. Range: _____
11. Interquartile range: _____
12. Variance: _____
13. Standard deviation: _____

Set C: 2, 30, 28, 5, 9, 20, 6, 11, 20, 9

14. Range: _____
15. Interquartile range: _____
16. Variance: _____
17. Standard deviation: _____

For the tables below, calculate the mode, median, mean, range, interquartile range, variance, and standard deviation.

18. Mode: _____
19. Median: _____
20. Mean: _____
21. Range: _____
22. Interquartile range: _____

Hours Per Day Watching TV

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	0	56	4.0	4.0	4.0
	1	311	22.0	22.0	25.9
	2	422	29.8	29.8	55.7
	3	282	19.9	19.9	75.6
	4	183	12.9	12.9	88.6
	5	95	6.7	6.7	95.3
	6	67	4.7	4.7	100.0
	Total	1416	100.0	100.0	

23. Variance: _____
24. Standard deviation: _____

Number of Brothers and Sisters

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	0	82	7.0	7.0	7.0
	1	238	20.3	20.3	27.2
	2	316	26.9	26.9	54.1
	3	264	22.5	22.5	76.6
	4	164	14.0	14.0	90.6
	5	111	9.4	9.4	100.0
	Total	1175	100.0	100.0	

25. Mode: _____
26. Median: _____
27. Mean: _____
28. Range: _____
29. Interquartile range: _____
30. Variance: _____
31. Standard deviation: _____

Age of Respondent

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	18	5	4.8	4.8	4.8
	19	17	16.2	16.2	21.0
	20	18	17.1	17.1	38.1
	21	22	21.0	21.0	59.0
	22	15	14.3	14.3	73.3
	23	28	26.7	26.7	100.0
	Total	105	100.0	100.0	

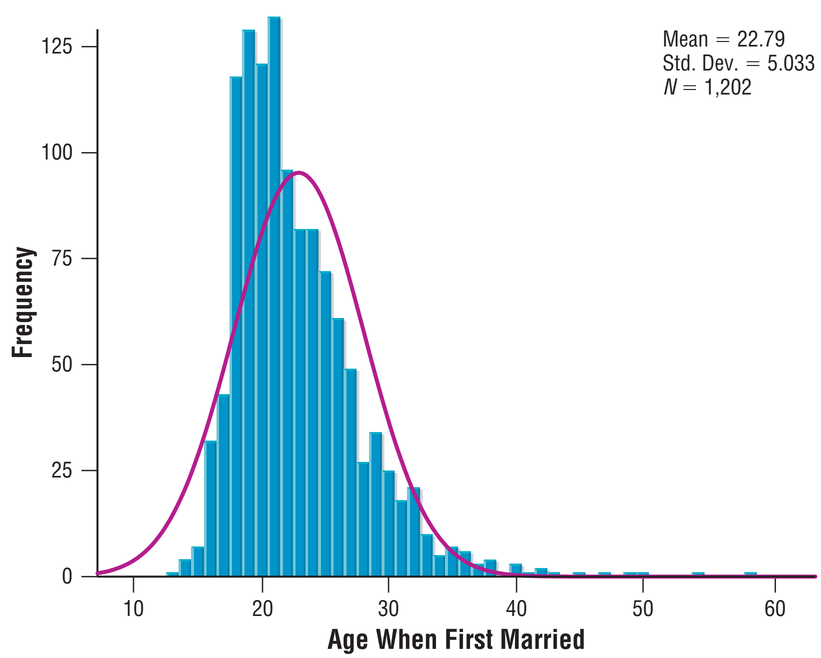
32. Mode: _____
33. Median: _____
34. Mean: _____
35. Range: _____
36. Interquartile range: _____
37. Variance: _____
38. Standard deviation: _____

Age When First Married

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	16	32	4.8	4.8	4.8
	17	43	6.4	6.4	11.2
	18	118	17.6	17.6	28.8
	19	129	19.2	19.2	48.0
	20	121	18.0	18.0	66.0
	21	132	19.7	19.7	85.7
	22	96	14.3	14.3	100.0
	Total	671	100.0	100.0	

39. Mode: _____
40. Median: _____
41. Mean: _____
42. Range: _____
43. Interquartile range: _____
44. Variance: _____
45. Standard deviation: _____

Use the chart below to answer the following questions.



46. What is the average age of respondents when they first married? _____

47. How many respondents provided data for this chart? _____

48. What is the variance equal to? _____

49. Based on the properties of the normal curve, we might predict that 68% of all respondents were married between the ages of _____ and _____.

50. Based on the properties of the normal curve, we might predict that 95% of all respondents were married between the ages of _____ and _____.

Computer Exercises

Use the GSS2006 database to answer the following questions. For some questions, you will need to know how to split the database into groups.

51. What is the standard deviation for age?

52. What is the standard deviation for age among males?

53. What is the standard deviation for age among females?

54. Which group is most dispersed around their respective average age: Whites, Blacks, or Others?

55. What is the range for the variable SEI (Socioeconomic Index)?

56. What is the standard deviation for SEI among males?

57. What is the standard deviation for SEI among females?

58. Based on our knowledge of the standard deviation we might predict that, among males, 68% of males will have SEI scores between _____ and _____.

59. Based on our knowledge of the standard deviation we might predict that, among females, 95% of females will have SEI scores between _____ and _____.

60. Using the mean and the standard deviation, compare the distribution of the number of years of education (EDUC) for males with that of females.

Now You Try It Answers

Answers to 1: Males: Range – 13 Mean deviation – 3.75
Females: Range – 5 Mean deviation – 1.5

Answers to 2: Mean – 7 Variance – 10.2 Standard deviation – 3.2
Answers to 3: Variance – 5.8 Standard deviation – 2.4

Key Terms

Box plot
Dispersion
Interquartile range
Mean deviation
Measures of dispersion
Normal curve
Range
Standard deviation
Variance

Works Cited

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Answers to Odd-Numbered Chapter Exercises and Computer Exercises

Exercises:

1. 6
3. 1.5
5. 2.41
7. 3
9. 1.89
11. 12
13. 6.82
15. 15
17. 9.84
19. 2
21. 6
23. 2.15
25. 2
27. 2.45
29. 2
31. 1.39
33. 21
35. 5
37. 2.48
39. 21
41. 19.56
43. 3
45. 1.68
47. 1202
49. 17.8 and 27.8

Computer Exercises:

51. 16.894
53. 17.229
55. 19.5995
57. 19.3104
59. 10.68 and 87.92